

Localized Model Order Reduction for Discrete-Ordinates Neutron Transport

F. A. B. Silva, J. C. Ragusa

TEXAS A&M UNIVERSITY
Department of Nuclear Engineering

Motivation

Setting:

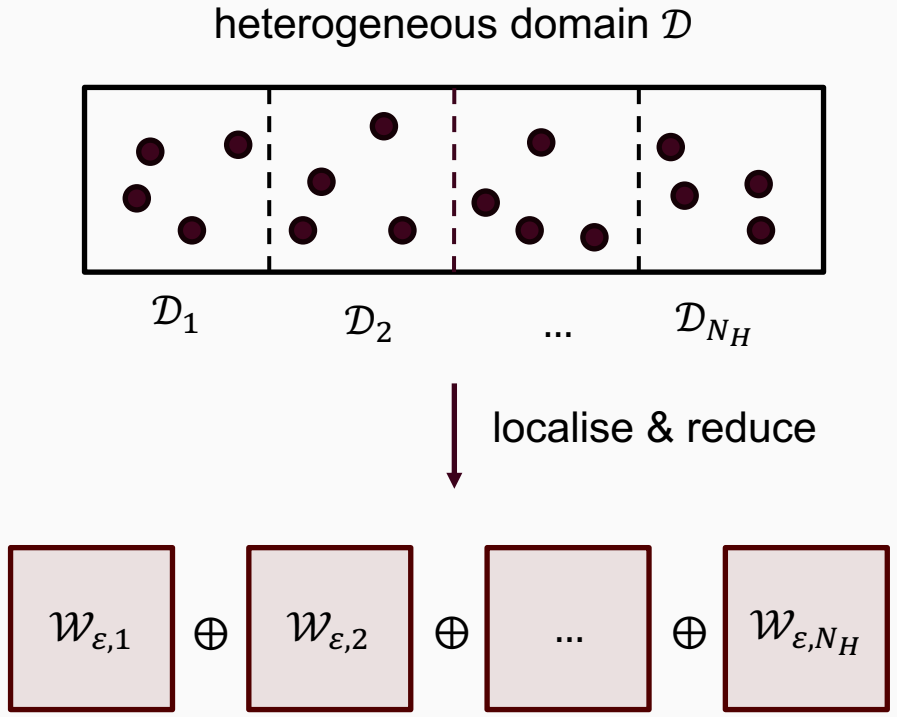
Stationary neutron transport on domains with many small inclusions or, in general, with parametrized heterogeneities.

Target:

Many-query problems where geometry and cross-sections vary independently per subdomain: a large dimensional parameter space in which *global* reduced basis is inefficient

Idea:

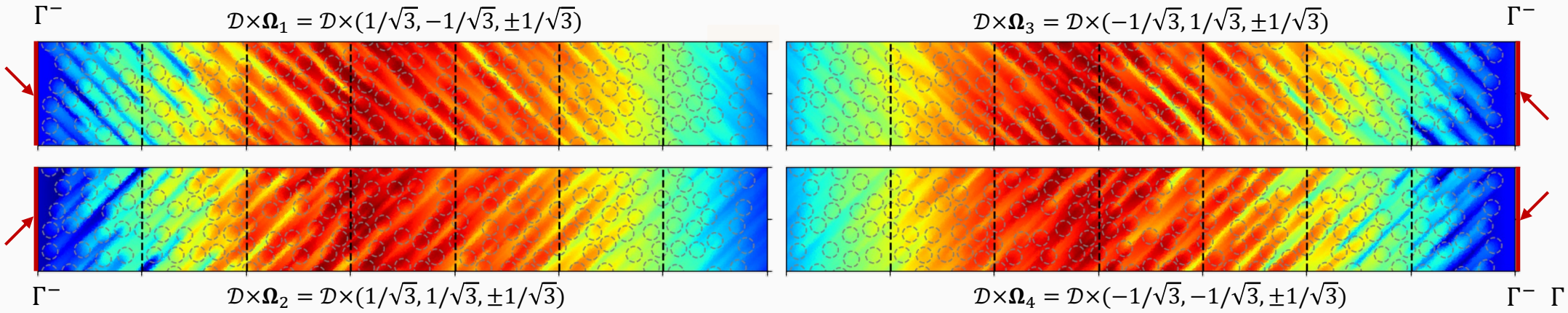
Build low-dimensional local spaces per subdomain, coupled through the DG numerical flux, i.e., build localized reduced bases (LRB).



The Neutron Transport Problem

Mono-energetic, steady state, heterogeneous (possibly multiplying) medium on the phase space $\mathcal{D}\times\mathcal{S}^2$, with inflow/outflow split $\Gamma^{\pm}$ and inflow data $\Psi(\mathbf{x},\boldsymbol{\Omega}) = \Psi^{\text{inc}}(\mathbf{x},\boldsymbol{\Omega})$ on Γ^{-}

$$\underbrace{\boldsymbol{\Omega}\cdot\nabla_{\mathbf{x}}\Psi}_{\text{streaming}} + \underbrace{\sigma_t\Psi}_{\text{collision}} - \underbrace{\frac{\sigma_s+\nu\sigma_f}{4\pi}\int_{\mathcal{S}^2}\Psi\,\mathrm{d}\boldsymbol{\Omega}'}_{\text{scattering + fission}} = \underbrace{\frac{Q}{4\pi}}_{\text{source}}$$



High-fidelity Model: Discrete Ordinates in Angle

Replace the angular variable by a quadrature on $\mathcal{S}^2$: $\{(\mathbf{\Omega}_\mathrm{d}, \omega_\mathrm{d})\}_{\mathrm{d}=1}^{N_\Omega}$, $\sum_{\mathrm{d}=1}^{N_\Omega} \omega_\mathrm{d} = 4\pi$ employing N_Ω discrete ordinates. For example, level symmetric or Lebdev.

The angular collocation turns the integro-differential equation into a coupled system of *advection-reaction* partial differential equations, one per direction:

$$\mathbf{\Omega}_\mathrm{d} \cdot \nabla_\mathbf{x} \Psi_\mathrm{d} + \sigma_t \Psi_\mathrm{d} - \frac{\sigma_s + \nu\sigma_f}{4\pi} \Phi = \frac{Q}{4\pi}, \qquad \Phi = \sum_{\mathrm{d}'=1}^{N_\Omega} \omega_{\mathrm{d}'} \Psi_{\mathrm{d}'}$$

- each equation has a constant transport direction $\mathbf{\Omega}_\mathrm{d}$
- directions couple through the scalar flux $\Phi = \sum_{\mathrm{d}'=1}^{N_\Omega} \omega_{\mathrm{d}'} \Psi_{\mathrm{d}'}$

High-fidelity Model: Upwind DG in Space

Discretization in space uses a broken piecewise-affine space $\mathcal{W}_h = \mathcal{V}_h(\mathcal{D})^{N_\Omega}$ on a mesh $\mathcal{T}_h$ resolving the material interfaces. Here we assume $\mathcal{V}_h(\mathcal{D}) = \{\varphi \in L^2(\mathcal{D}) : \varphi|_K \in \mathbb{P}^n(K)\ \forall K \in \mathcal{T}_h\}$ and $\dim \mathcal{V}_h(\mathcal{D}) = N_h \longrightarrow \mathcal{N} = N_\Omega N_h$

Employing a stable upwind DG scheme, allows to naturally handle $\sigma_t, \sigma_s, \sigma_f$ and Q discintinuities across interfaces. The upwind numerical flux is defined by

$$\hat{\mathrm{u}}_\mathrm{d} = \begin{cases} \mathrm{u}_d|_E\ , & \boldsymbol{\Omega}_\mathrm{d} \cdot \mathbf{n}_E \geq 0 \quad \text{outflow: own value,} \\ \mathrm{u}_d|_{E'}\ , & \boldsymbol{\Omega}_\mathrm{d} \cdot \mathbf{n}_E < 0 \quad \text{inflow: upwind neighbor,} \\ 0\ , & \boldsymbol{\Omega}_\mathrm{d} \cdot \mathbf{n}_E < 0 \quad \text{inflow: on } \partial\mathcal{D}. \end{cases}$$

where E is the current element/subdomain, E' its neighbor across the face, and $\mathbf{n}_E$ the outward normal.

Variational Formulation

The discrete weak problem can be written in terms of linear and bilinear forms, introducing ℓ_h and $a_h = t_h - b_h$. Here $t_h, b_h : \mathcal{W}_h \times \mathcal{W}_h \to \mathbb{R}$ are the *transport* and the *scattering-fission* bilinear forms, defined by

$$t_h(\mathbf{u},\mathbf{v}) = \sum_{\mathrm{d}=1}^{N_\Omega} \omega_\mathrm{d} \sum_{K\in\mathcal{T}_h} \int_K \sigma_t \mathrm{u}_\mathrm{d} \mathrm{v}_\mathrm{d} d\mathbf{x} - \int_K \mathrm{u}_\mathrm{d} (\mathbf{\Omega}_\mathrm{d} \cdot \nabla_\mathbf{x} \mathrm{v}_\mathrm{d}) d\mathbf{x} + \int_{\partial K} (\mathbf{\Omega}_\mathrm{d} \cdot \mathbf{n}_E) \hat{\mathrm{u}}_\mathrm{d} \mathrm{v}_\mathrm{d} dS(\mathbf{x})$$

$$b_h(\mathbf{u},\mathbf{v}) = \int_\mathcal{D} \frac{\sigma_s + \nu\sigma_f}{4\pi} \Phi_\mathrm{u} \Phi_\mathrm{v} d\mathbf{x}, \qquad \Phi_\mathrm{u} = \sum_{\mathrm{d}'=1}^{N_\Omega} \omega_{\mathrm{d}'} \mathrm{u}_{\mathrm{d}'},\ \Phi_\mathrm{v} = \sum_{\mathrm{d}'=1}^{N_\Omega} \omega_{\mathrm{d}'} \mathrm{v}_{\mathrm{d}'}$$

and $\ell_h : \mathcal{W}_h \to \mathbb{R}$ is the *load* functional, including the source and the prescribed inflow terms

$$\ell_h(\mathbf{v}) = \sum_{\mathrm{d}=1}^{N_\Omega} \omega_\mathrm{d} \int_\mathcal{D} \frac{Q}{4\pi} \mathrm{v}_\mathrm{d} d\mathbf{x} + \sum_{\mathrm{d}=1}^{N_\Omega} \omega_\mathrm{d} \int_{\partial_\mathrm{d}^- \mathcal{D}} |\mathbf{\Omega}_\mathrm{d} \cdot \mathbf{n}_E| \, \Psi^\mathrm{inc} \mathrm{v}_\mathrm{d} dS(\mathbf{x}), \qquad \partial_\mathrm{d}^- \mathcal{D} = \{\mathbf{x} \in \partial\mathcal{D} | \, \mathbf{\Omega}_\mathrm{d} \cdot \mathbf{n}_E(\mathbf{x}) < 0\}$$

Full order Sn-DG problem:

Find $\mathbf{\Psi}_h = \{\Psi_{h,\mathrm{d}}\}_{\mathrm{d}=1}^{N_\Omega} \in \mathcal{W}_h$ such that $a_h(\mathbf{\Psi}_h, \mathbf{v}_h) = \ell_h(\mathbf{v}_h)$ for all $\mathbf{v}_h \in \mathcal{W}_h$.

From Global to Local

The passage from *global* to *local* is obtained replacing the global high-fidelity space $\mathcal{W}_h$, with the direct a sum of local reduced basis spaces

$$\mathcal{W}_\varepsilon = \bigoplus_{i=1}^{N_H} \mathcal{W}_{\varepsilon,i} \subset \mathcal{W}_h \qquad \dim \mathcal{W}_\varepsilon = \sum_{i=1}^{N_H} \dim \mathcal{W}_{\varepsilon,i} = \sum_{i=1}^{N_H} N_{\varepsilon,i} \ll \mathcal{N}$$

- each $\mathcal{W}_{\varepsilon,i}$ must capture the transport dynamics, the local heterogeneities and the coupling to neighbors
- coupling between subdomains happens through the upwind flux only, making the reduced problem an exact Galerkin restriction of the high-fidelity scheme

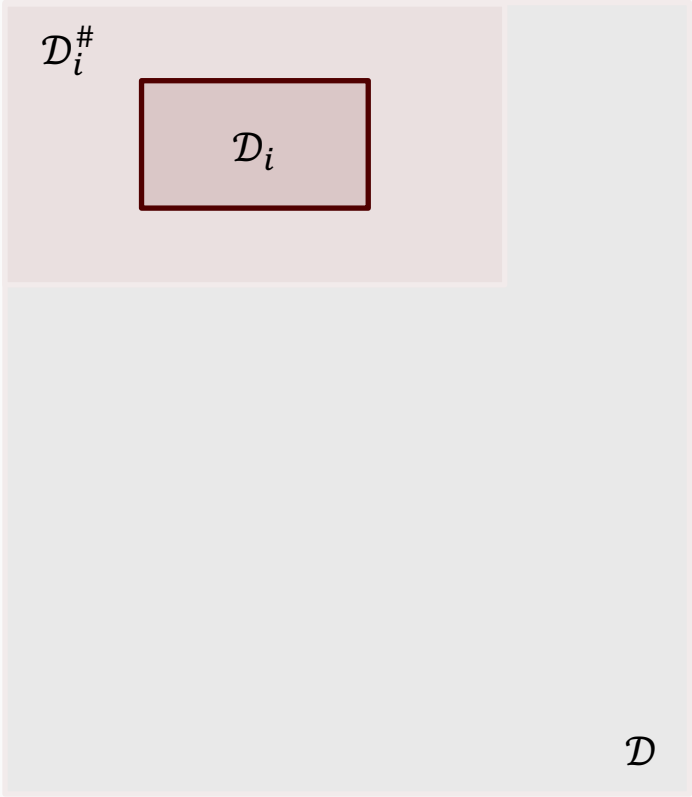
Two questions

(i) How do we construct good approximation spaces? **(ii)** Can we certify the error?

The *optimal local approximation* theory from Babuška–Lipton and Smetana–Patera, is here adopted to a *non-self-adjoint, non-coercive* transport setting.

Coarse Decomposition and Oversampling

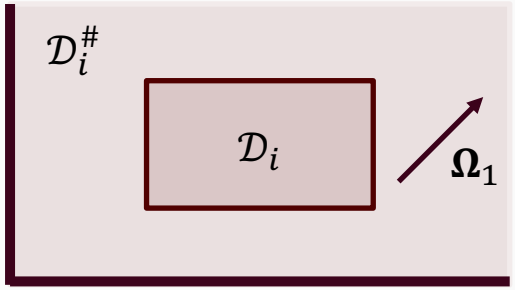
- given the non-overlapping coarse partition $\{\mathcal{D}_i\}_{i=1}^{N_H}$, the oversampling regions $\mathcal{D} \supset \mathcal{D}_i^{\#} \supset \mathcal{D}_i$ are selected to buffer the artificial boundary effects



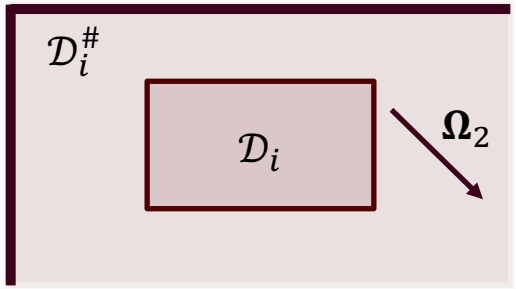
Coarse Decomposition and Oversampling

- given the non-overlapping coarse partition $\{\mathcal{D}_i\}_{i=1}^{N_H}$, the oversampling regions $\mathcal{D} \supset \mathcal{D}_i^{\#} \supset \mathcal{D}_i$ are selected to buffer the artificial boundary effects

- inflow boundaries are *per ordinate,* rotating with $\boldsymbol{\Omega}_{\mathrm{d}}$, specifically, $\partial_{\mathrm{d}}^{-} E = \{\mathbf{x} \in \partial E \mid \boldsymbol{\Omega}_{\mathrm{d}} \cdot \mathbf{n}_E(\mathbf{x}) < 0\}$



$\boldsymbol{\Omega}_1 = (1/\sqrt{3}, 1/\sqrt{3})$: inflow = left + bottom



$\boldsymbol{\Omega}_2 = (1/\sqrt{3}, -1/\sqrt{3})$: inflow = left + top

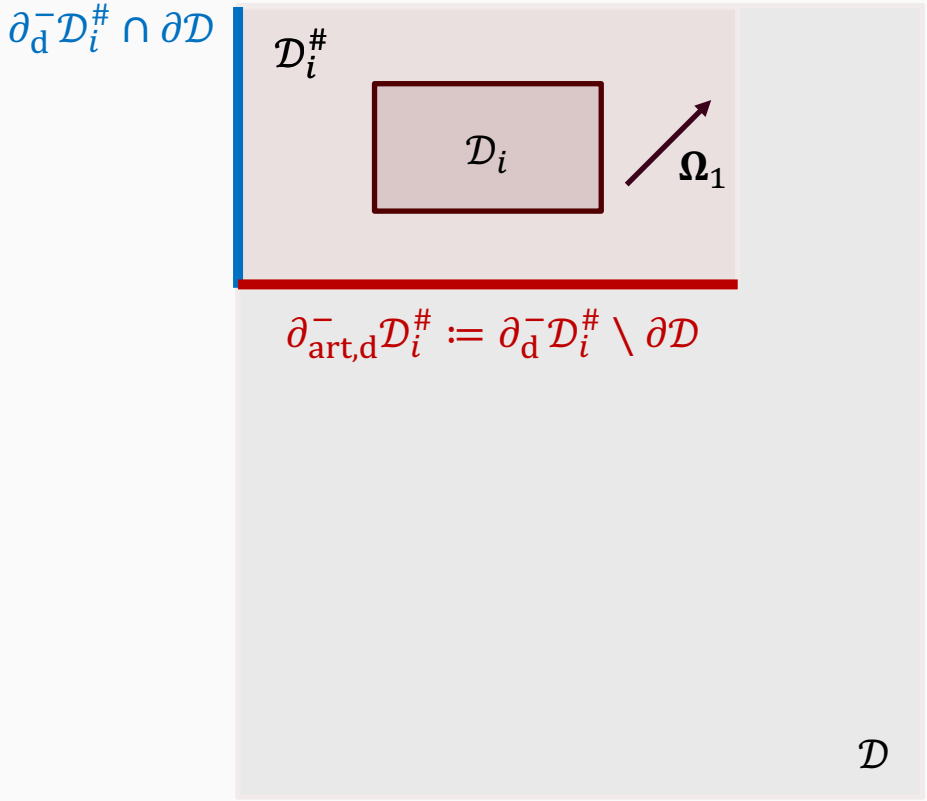
— inflow boundary $\partial_{\mathrm{d}}^{-} \mathcal{D}_i^{\#}$

Coarse Decomposition and Oversampling

- given the non-overlapping coarse partition $\{\mathcal{D}_i\}_{i=1}^{N_H}$, the oversampling regions $\mathcal{D} \supset \mathcal{D}_i^{\#} \supset \mathcal{D}_i$ are selected to buffer the artificial boundary effects

- inflow boundaries are *per ordinate,* rotating with $\boldsymbol{\Omega}_\mathrm{d}$, specifically, $\partial_\mathrm{d}^- E = \{\mathbf{x} \in \partial E \mid \boldsymbol{\Omega}_\mathrm{d} \cdot \mathbf{n}_E(\mathbf{x}) < 0\}$

- inflow boundaries of $\mathcal{D}_i^{\#}$ split into *physical* b. (on $\partial\mathcal{D}$), carrying the boundary data Ψ^{inc}, and *artificial* b. ($\boldsymbol{\partial}_{\mathrm{art}}^- \mathcal{D}_i^{\#}$), carrying transmission data



Coarse Decomposition and Oversampling

- given the non-overlapping coarse partition $\{\mathcal{D}_i\}_{i=1}^{N_H}$, the oversampling regions $\mathcal{D} \supset \mathcal{D}_i^{\#} \supset \mathcal{D}_i$ are selected to buffer the artificial boundary effects

- inflow boundaries are *per ordinate,* rotating with $\boldsymbol{\Omega}_\mathrm{d}$, specifically, $\partial_\mathrm{d}^- E = \{\mathbf{x} \in \partial E \mid \boldsymbol{\Omega}_\mathrm{d} \cdot \mathbf{n}_E(\mathbf{x}) < 0\}$

- inflow boundaries of $\mathcal{D}_i^{\#}$ split into *physical* b. (on $\partial\mathcal{D}$), carrying the boundary data Ψ^{inc}, and *artificial* b. ($\boldsymbol{\partial}_{\mathrm{art}}^- \mathcal{D}_i^{\#}$), carrying transmission data

Finally, we introduce the transport-weighted inflow-trace inner product:

$$(\mathbf{u}, \mathbf{v})_{\partial_\mathrm{d}^- E} = \sum_{\mathrm{d}=1}^{N_\Omega} \omega_\mathrm{d} \int_{\partial_\mathrm{d}^- E} |\boldsymbol{\Omega}_\mathrm{d} \cdot \mathbf{n}_E| \, \mathrm{u}_\mathrm{d} \mathrm{v}_\mathrm{d} dS(\mathbf{x})$$

The Local Transfer Operators

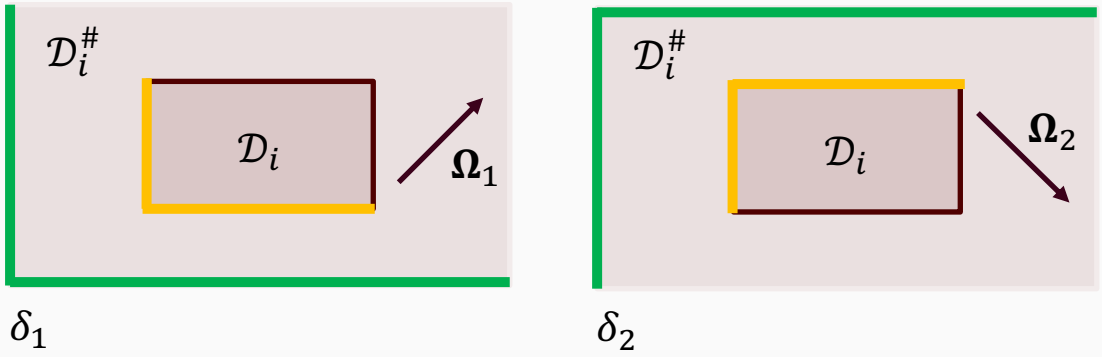
Consider the homogenous local lift $\boldsymbol{\Psi}_{h,i}^{\mathrm{hom}} \in \mathcal{W}_h(\mathcal{D}_i^{\#})$ mapping the inflow-trace $\boldsymbol{\delta} \in \mathcal{W}_h(\boldsymbol{\partial}_{\mathrm{art}}^{-}\mathcal{D}_i^{\#})$ to a space-angle state defined on the full oversampling region. The state solves the following Sn–DG problem:

$$-\int_K \Psi_{h,i,\mathrm{d}}^{\mathrm{hom}}\, \boldsymbol{\Omega}_{\mathrm{d}} \cdot \nabla_{\mathbf{x}} \zeta_h d\mathbf{x} + \int_{\partial K} (\boldsymbol{\Omega}_{\mathrm{d}} \cdot \mathbf{n}_K)\, \widehat{\Psi}_{h,i,\mathrm{d}}^{\mathrm{hom}}\, \zeta_h dS(\mathbf{x}) + \int_K \sigma_t\, \Psi_{h,i,\mathrm{d}}^{\mathrm{hom}} \zeta_h d\mathbf{x} - \int_K \frac{\sigma_s + \nu\sigma_f}{4\pi} \Phi_{h,i}^{\mathrm{hom}} \zeta_h d\mathbf{x} = 0,$$

with $\Phi_{h,i}^{\mathrm{hom}} = \sum_{\mathrm{d}=1}^{N_{-}\Omega} \omega_{\mathrm{d}} \Psi_{h,i,\mathrm{d}}^{\mathrm{hom}}$, zero source, zero physical inflow, and $\widehat{\Psi}_{h,i,\mathrm{d}}^{\mathrm{hom}}|_{\boldsymbol{\partial}_{\mathrm{art}}^{-}\mathcal{D}_i^{\#}} = \boldsymbol{\delta}$.

Boundary transfer map: $\mathcal{G}_i$

$\mathcal{G}_i \boldsymbol{\delta} = \widehat{\Psi}_{h,i,\mathrm{d}}^{\mathrm{hom}}|_{\boldsymbol{\partial}_{\mathrm{art}}^{-}\mathcal{D}_i}$ (inflow trace on the target)

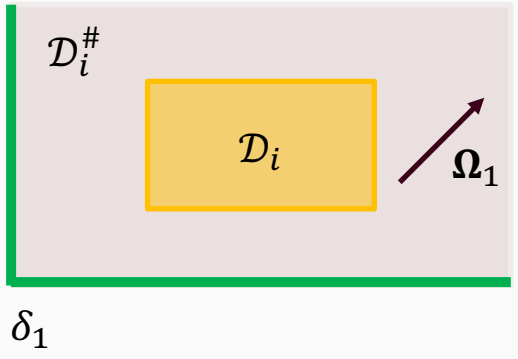


The Local Transfer Operators

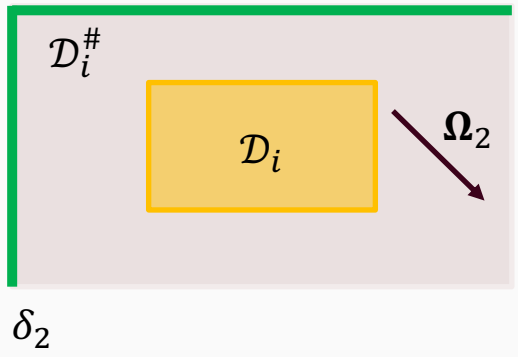
Consider the homogenous local lift $\boldsymbol{\Psi}_{h,i}^{\mathrm{hom}} \in \mathcal{W}_h(\mathcal{D}_i^{\#})$ mapping the inflow-trace $\boldsymbol{\delta} \in \mathcal{W}_h(\boldsymbol{\partial}_{\mathrm{art}}^{-}\mathcal{D}_i^{\#})$ to a space-angle state defined on the full oversampling region. The state solves the following Sn–DG problem:

$$-\int_K \Psi_{h,i,\mathrm{d}}^{\mathrm{hom}}\, \boldsymbol{\Omega}_{\mathrm{d}} \cdot \nabla_{\mathbf{x}} \zeta_h d\mathbf{x} + \int_{\partial K} (\boldsymbol{\Omega}_{\mathrm{d}} \cdot \mathbf{n}_K)\, \widehat{\Psi}_{h,i,\mathrm{d}}^{\mathrm{hom}}\, \zeta_h dS(\mathbf{x}) + \int_K \sigma_t\, \Psi_{h,i,\mathrm{d}}^{\mathrm{hom}} \zeta_h d\mathbf{x} - \int_K \frac{\sigma_s + \nu\sigma_f}{4\pi} \Phi_{h,i}^{\mathrm{hom}} \zeta_h d\mathbf{x} = 0,$$

with $\Phi_{h,i}^{\mathrm{hom}} = \sum_{\mathrm{d}=1}^{N_\Omega} \omega_{\mathrm{d}} \Psi_{h,i,\mathrm{d}}^{\mathrm{hom}}$, zero source, zero physical inflow, and $\widehat{\Psi}_{h,i,\mathrm{d}}^{\mathrm{hom}}|_{\boldsymbol{\partial}_{\mathrm{art}}^{-}\mathcal{D}_i^{\#}} = \boldsymbol{\delta}$.



δ_1



δ_2

Volume transfer map: $\mathcal{H}_i$

$\mathcal{H}_i \boldsymbol{\delta} = \widehat{\Psi}_{h,i,\mathrm{d}}^{\mathrm{hom}}|_{\mathcal{D}_i}$ (angular flux inside $\mathcal{D}_i$)

The Local Transfer Operators

Consider the homogenous local lift $\boldsymbol{\Psi}_{h,i}^{\mathrm{hom}} \in \mathcal{W}_h(\mathcal{D}_i^{\#})$ mapping the inflow-trace $\boldsymbol{\delta} \in \mathcal{W}_h(\boldsymbol{\partial}_{\mathrm{art}}^{-}\mathcal{D}_i^{\#})$ to a space-angle state defined on the full oversampling region. The state solves the following Sn–DG problem:

$$-\int_K \Psi_{h,i,\mathrm{d}}^{\mathrm{hom}}\, \boldsymbol{\Omega}_{\mathrm{d}} \cdot \nabla_{\mathbf{x}} \zeta_h d\mathbf{x} + \int_{\partial K} (\boldsymbol{\Omega}_{\mathrm{d}} \cdot \mathbf{n}_K)\, \widehat{\Psi}_{h,i,\mathrm{d}}^{\mathrm{hom}}\, \zeta_h dS(\mathbf{x}) + \int_K \sigma_t\, \Psi_{h,i,\mathrm{d}}^{\mathrm{hom}} \zeta_h d\mathbf{x} - \int_K \frac{\sigma_s + \nu\sigma_f}{4\pi}\, \Phi_{h,i}^{\mathrm{hom}} \zeta_h d\mathbf{x} = 0,$$

with $\Phi_{h,i}^{\mathrm{hom}} = \sum_{\mathrm{d}=1}^{N_\Omega} \omega_{\mathrm{d}} \Psi_{h,i,\mathrm{d}}^{\mathrm{hom}}$, zero source, zero physical inflow, and $\widehat{\Psi}_{h,i,\mathrm{d}}^{\mathrm{hom}}|_{\boldsymbol{\partial}_{\mathrm{art}}^{-}\mathcal{D}_i^{\#}} = \boldsymbol{\delta}$.

Boundary transfer map: $\mathcal{G}_i$

$\mathcal{G}_i \boldsymbol{\delta} = \widehat{\Psi}_{h,i,\mathrm{d}}^{\mathrm{hom}}|_{\boldsymbol{\partial}_{\mathrm{art}}^{-}\mathcal{D}_i}$ (inflow trace on the target)

Volume transfer map: $\mathcal{H}_i$

$\mathcal{H}_i \boldsymbol{\delta} = \widehat{\Psi}_{h,i,\mathrm{d}}^{\mathrm{hom}}|_{\mathcal{D}_i}$ (angular flux inside $\mathcal{D}_i$)

The spectrum of $\mathcal{G}_i$ is analyzed to extract optimal boundary modes; $\mathcal{H}_i$ is used to transport them into the volume.

Optimal Local Approximation Spaces

The worst-case boundary transfer error on a boundary space $\tilde{\mathcal{U}}$ is:

$$d_{\mathcal{G}_i}(\tilde{\mathcal{U}}) = \sup_{\boldsymbol{\delta}\neq 0} \inf_{\boldsymbol{u}\in\tilde{\mathcal{U}}} \frac{\|\mathcal{G}_i\boldsymbol{\delta} - \boldsymbol{u}\|_{\partial^-_{\mathrm{art}}\mathcal{D}_i}}{\|\boldsymbol{\delta}\|_{\partial^-_{\mathrm{art}}\mathcal{D}_i^\#}}.$$

The Kolmogorov-optimal space minimizing this distance is spanned by the dominant transfer modes obtained from the singular value decomposition of $\mathcal{G}_i$ or, equivalently, from the eigenvalue problem for $\mathcal{G}_i^*\mathcal{G}_i$

$$\left(\mathcal{G}_i\boldsymbol{\delta}_{i,j}, \mathcal{G}_i\boldsymbol{\eta}\right)_{\partial^-_{\mathrm{art}}\mathcal{D}_i} = \lambda_{i,j}\left(\boldsymbol{\delta}_{i,j}, \boldsymbol{\eta}\right)_{\partial^-_{\mathrm{art}}\mathcal{D}_i^\#}, \qquad \lambda_{i,1} \geq \lambda_{i,2} \geq \cdots \geq 0,$$

the optimal error is given by $\sqrt{\lambda_{i,N_{\varepsilon,i}+1}}$, i.e., the first discarded singular value.

Homogenous modes

$\boldsymbol{\chi}_{i,j}^{\mathrm{hom}} = \mathcal{H}_i\boldsymbol{\delta}_{i,j}$ maps the boundary modes to the homogeneous volume basis functions.

Source and Physical Inflow: Particular Modes

Homogeneous modes described so far carry only *inter-subdomain transmission.* We still need the response to the volumetric source Q and to the physical inflow Ψ^{inc}. For this purpose, consider the Sn-DG problem:

$$
\begin{aligned}
&-\int_K \Psi^{\text{par}}_{h,i,\text{d}}\, \boldsymbol{\Omega}_\text{d}\cdot\nabla_{\mathbf{x}}\zeta_h d\mathbf{x} + \int_{\partial K}(\boldsymbol{\Omega}_\text{d}\cdot\mathbf{n}_K)\,\widehat{\Psi}^{\text{par}}_{h,i,\text{d}}\,\zeta_h dS(\mathbf{x}) + \int_K \sigma_t\,\Psi^{\text{par}}_{h,i,\text{d}}\zeta_h d\mathbf{x} - \int_K \frac{\sigma_s+\nu\sigma_f}{4\pi}\Phi^{\text{par}}_{h,i}\zeta_h d\mathbf{x}\\
&= \int_K \frac{Q}{4\pi}\zeta_h d\mathbf{x} + \int_{\partial^-_\text{d}\mathcal{D}\cap\partial K} |\boldsymbol{\Omega}_\text{d}\cdot\mathbf{n}_E|\,\Psi^{\text{inc}}\zeta_h dS(\mathbf{x}),
\end{aligned}
$$

with $\Phi^{\text{par}}_{h,i} = \sum_{\text{d}=1}^{N_\Omega}\omega_\text{d}\Psi^{\text{par}}_{h,i,\text{d}}$, and zero artificial inflow boundary conditions, i.e., $\widehat{\Psi}^{\text{par}}_{h,i,\text{d}}|_{\partial^-_{\text{art}}\mathcal{D}_i^\#} = \mathbf{0}$.

Particular data response mode

$\boldsymbol{\chi}^{\text{par}}_i = \widehat{\Psi}^{\text{par}}_{h,i,\text{d}}|_{\mathcal{D}_i}$ defines the local lift on $\mathcal{D}_i$ of the actual source and physical inflow data.

The Coarse-grid Reduced Problem

Assemble the local reduced spaces: $\mathcal{W}_{\varepsilon,i} = \operatorname{span}\{\boldsymbol{\chi}_{i,j}^{\mathrm{hom}}\} \oplus \operatorname{span}\{\boldsymbol{\chi}_{i}^{\mathrm{par}}\}$, for every $i \in N_H$.

Gluing together the local spaces without continuity gives $\mathcal{W}_\varepsilon \subset \mathcal{W}_h$. Consequently, the reduced problem is the Galerkin restriction of the full-order problem with the same a_h and ℓ_h:

Reduced problem

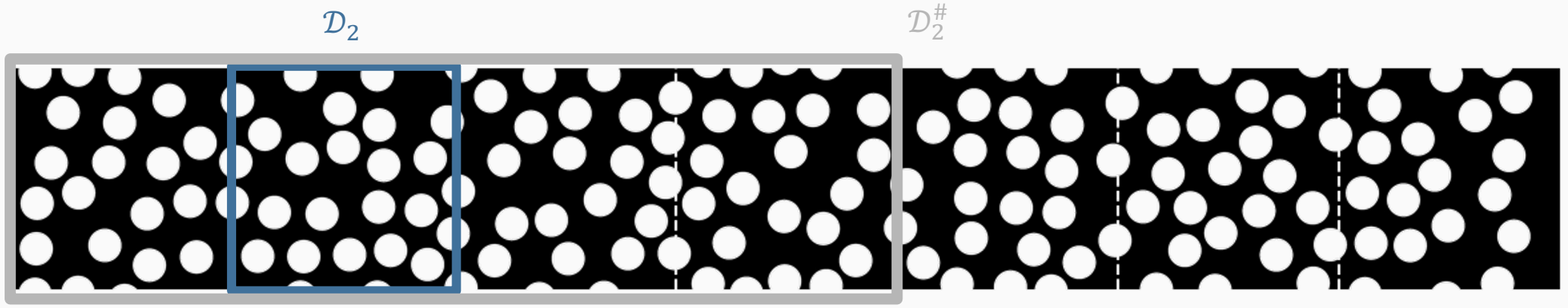
Find $\boldsymbol{\Psi}_\varepsilon = \{\Psi_{\varepsilon,\mathrm{d}}\}_{\mathrm{d}=1}^{N_\Omega} \in \mathcal{W}_\varepsilon$ such that $a_h(\boldsymbol{\Psi}_\varepsilon, \boldsymbol{\xi}) = \ell_h(\boldsymbol{\xi})$ for all $\boldsymbol{\xi} \in \mathcal{W}_\varepsilon$.

Testing subdomain-by-subdomain ($\boldsymbol{\xi}_\varepsilon \in \mathcal{W}_{\varepsilon,i}$) makes the upwind coupling explicit:

$$\sum_{\mathrm{d}=1}^{N_\Omega} \omega_{\mathrm{d}} \left[\sum_{K \in \mathcal{D}_i} \underbrace{\int_{\partial K \setminus \partial \mathcal{D}_i} (\boldsymbol{\Omega}_{\mathrm{d}} \cdot \mathbf{n}_K) \widehat{\Psi}_{\varepsilon,\mathrm{d}} \xi_{\mathrm{d}} \, dS(\mathbf{x})}_{\text{intra: fine faces in } \partial \mathcal{D}_i} + \underbrace{\int_{\partial \mathcal{D}_i} (\boldsymbol{\Omega}_{\mathrm{d}} \cdot \mathbf{n}_{\mathcal{D}_i}) \widehat{\Psi}_{\varepsilon,\mathrm{d}} \xi_{\mathrm{d}} \, dS(\mathbf{x})}_{\text{inter: coarse skeleton}} + \underbrace{\int_{\mathcal{D}_i} \sigma_t \Psi_{\varepsilon,\mathrm{d}} \xi_{\mathrm{d}} - \Psi_{\varepsilon,\mathrm{d}} (\boldsymbol{\Omega}_{\mathrm{d}} \cdot \nabla_{\mathbf{x}} \xi_{\mathrm{d}}) + \cdots}_{\text{volume}} \right]$$

On the skeleton the upwind trace draws the neighbor's value on inflow. No separate transmission condition needed.

A Heterogenous Benchmark



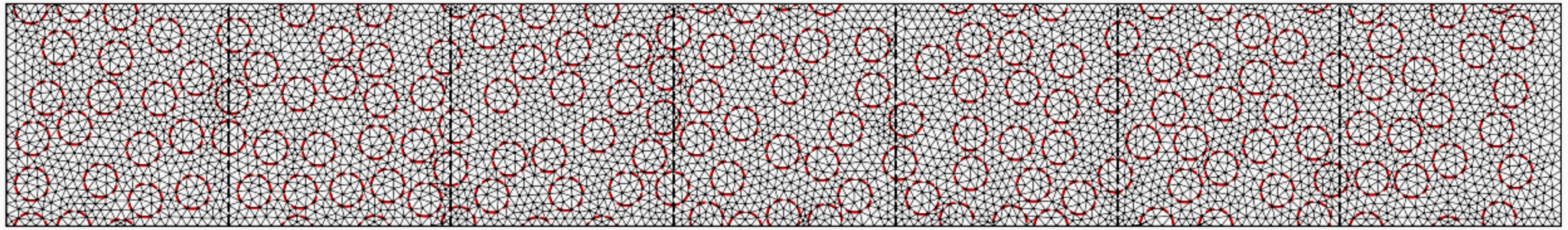
- $\mathcal{D}=[0,7]\times[0,1]$, $N_H=7$ unit subdomains with oversampling $R=2$
- 140 non overlapping, reactive, circular inclusions
- top and bottom boundaries are periodically identified
- constant inflow Ψ^{inc} on Γ^- and $Q=0$
- $N_\Omega=4$ directions, $N_{\mathbf{x}}=91{,}284$ for a total of $\mathcal{N}=365{,}136$

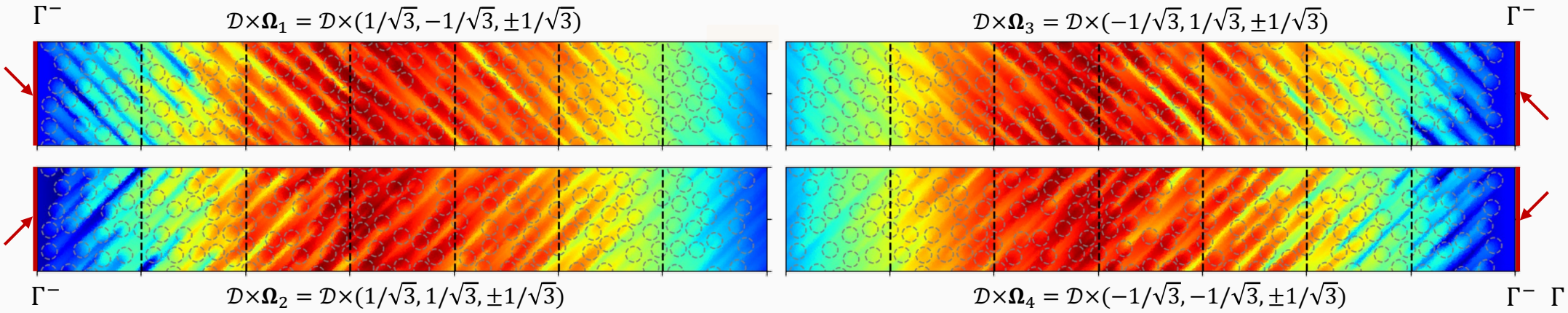
	background	inclusions
σ_t	0.1.	7.0
$\sigma_s+\nu\sigma_f$	0.0	7.2

inclusions are locally multiplying

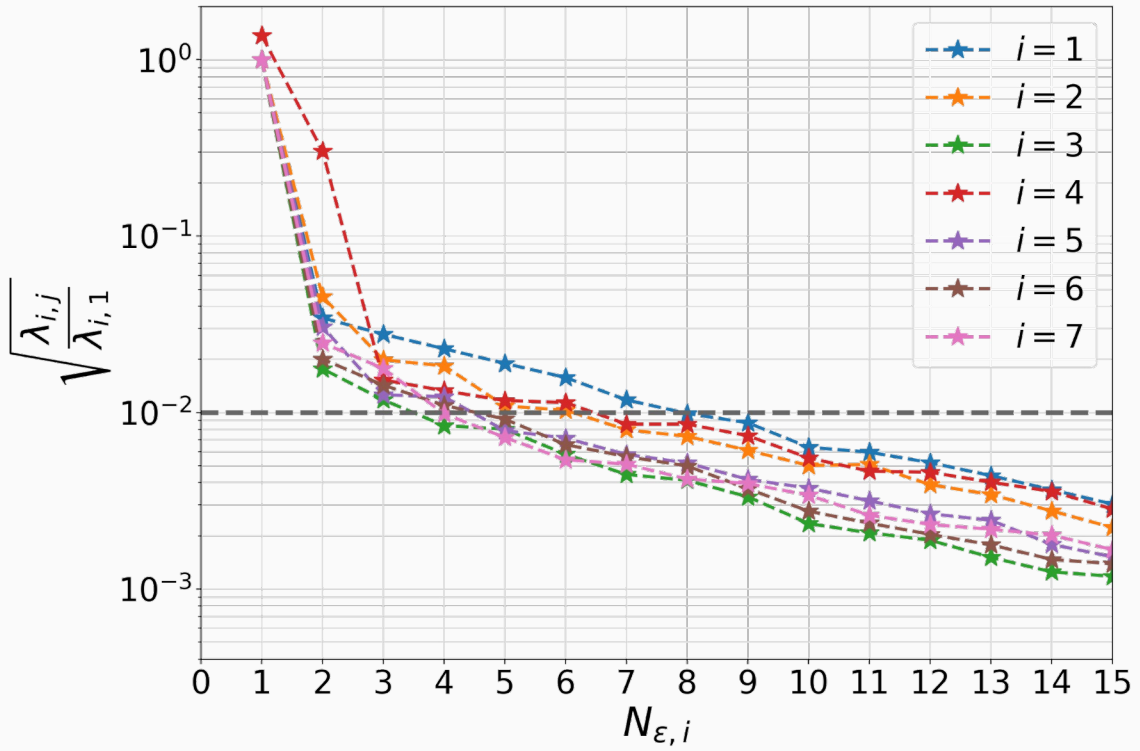
Full Order Reference Solution

fine DG triangulation resolving inclusions and coarse interface





Decay of the Local Transfer Spectra

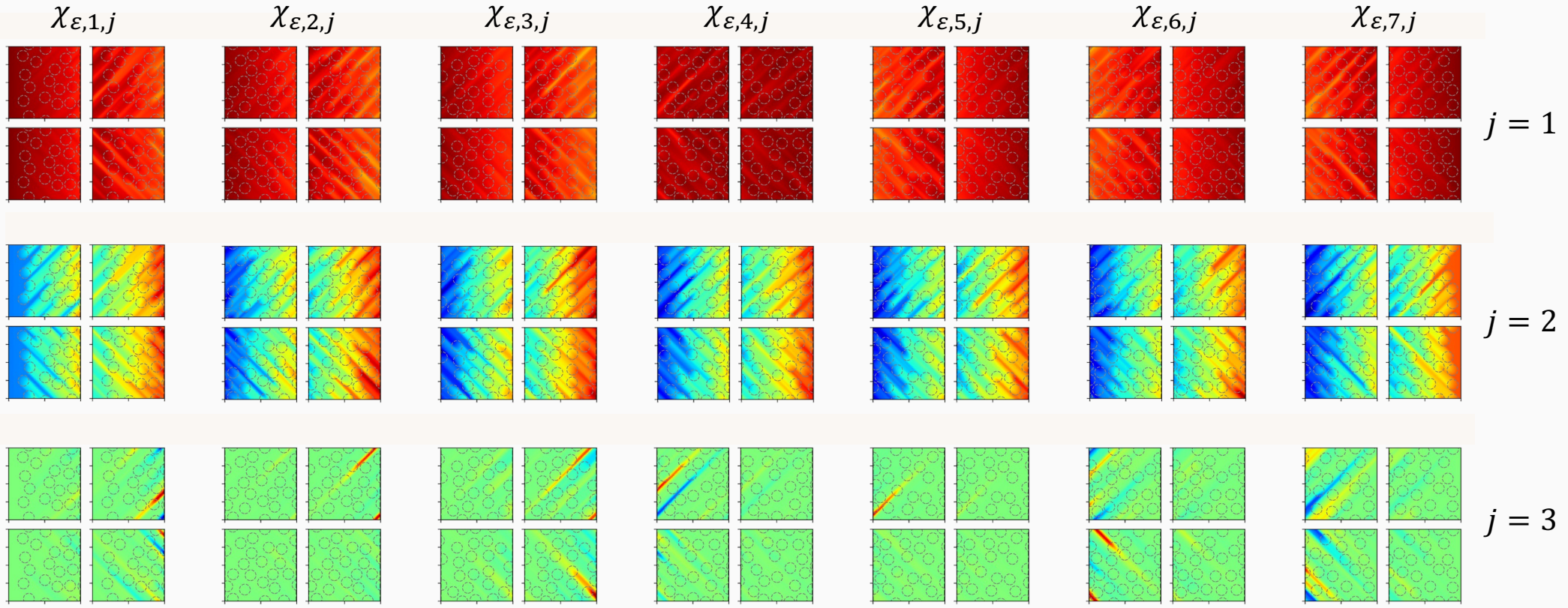


- singular values decay on all seven subdomains, few local mode suffice for an accurate reconstruction
- decay are slightly non uniform: more heterogenous and directional buffers decay more slowly

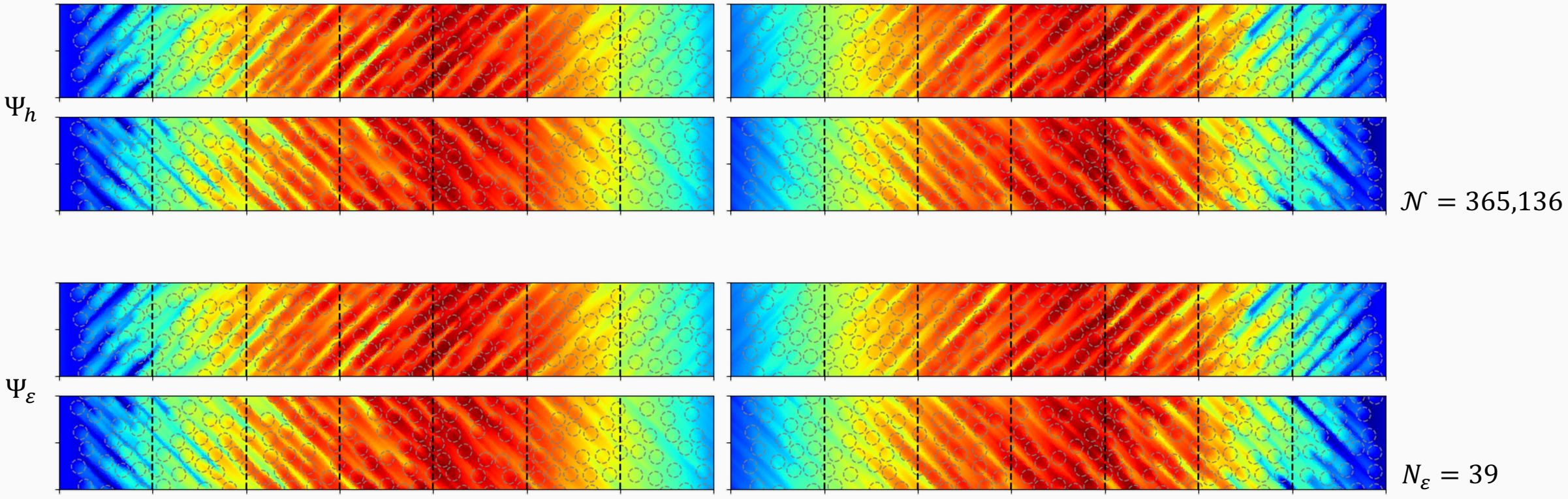
Working conjecture for the coercive regime

$$\sqrt{\lambda_{i,N_{\varepsilon,i}}} \lesssim C_M e^{-c_M \mathrm{Kn}} + C_D e^{-c_V (j-1)^{1/(d-1)}}$$

Optimal Local Basis



Localized Reduced Basis Solution



relative error $= 1.9127 \times 10^{-4}$

$\times 661$ online speed-up

Summary and Outlook

What we have

- optimal local approximation expanded to non-coercive Sn-DG transport
- local spaces coupled purely by the upwind fluxes, and exact Galerkin restriction
- A priori (transfer spectra) and a posteriori (data + skeleton) error control
- Benchmark with a clear transport spectra decay in strongly heterogeneous, locally multiplying media

Where we are going

- multi-group energy dependence
- model-free basis generation for random geometry
- parametric cross sections varying per subdomain
- parameter-robust local spaces via spectral-greedy

From local to global: modular reduced models for transport in random heterogeneous media.

Selected References

[1] K. Smetana, A. T. Patera, *Optimal local approximation spaces for component-based static condensation procedures,* SIAM Journal on Scientific Computing, 2016

[2] I. Babuška, R. Lipton, *Optimal local approximation spaces for generalized FEM*, Multiscale Modeling and Simulation, 2011

[3] A. Buhr, K. Smetana, *Randomized local model order reduction*, SIAM Journal on Scientific Computing, 2018

[4] D. B. P. Huynh, D, J. Knezevic, and A. T. Patera, *A static condensation reduced basis element method,* ESAIM: M2AN, 2013

[5] Y. Efendiev, J. Galvis, T. Y. Hou, *Generalized multiscale finite element methods (GMsFEM)* Journal of computational physics, 2013

[6] A. Abdulle, E. Weinan, B. Engquist, E. Vanden-Eijnden, *The heterogeneous multiscale method,* Acta Numerica, 2012

[7] N. Halko, P.-G. Martinsson, J. A. Tropp, *Finding structure in randomness*, SIAM Review, 2011



THANK YOU

Texas A&M University
Department of Nuclear Engineering

francesco.silva@tamu.edu